

Namita Kukreja

Software Development Engineer II

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SUMMARY

SDE-II with 7+ years of backend engineering experience across **microservices, distributed systems, and REST API design**, complemented by frontend work in ReactJS/Redux. Currently specialized in AI and LLM-powered tooling, architecting **agentic systems, RAG pipelines, and vector database** integrations at Autodesk, including Cursor/Claude MCP workflows. Proven track record of **platform-scale delivery** across Autodesk, Dassault Systèmes, and Persistent Systems.

WORK EXPERIENCE

Autodesk | Pune, India

Aug 2022 – Present

Software Development Engineer II

Stack: Node.js · Python · AWS · Microservices · Agentic AI

Client-Side SDK Architecture

- Designed and delivered a layered SDK abstraction over Autodesk's design platform.
- Enforced **cross-layer data integrity** and controlled metadata propagation across storage and retrieval services.
- Exposed clean, **versioned developer-facing APIs** over complex internal engineering models.

APS JSON Protocol Migration

- Architected migration of legacy XML-based changeset formats to a lightweight APS JSON exchange protocol.
- Reduced average API payload from ~850 KB to ~510 KB (**~40% reduction**).
- Delivered measurable **latency improvements** across Autodesk's design ecosystem.

OneGraph Query Microservice

- Designed and shipped a standalone filtered-read microservice over large-scale asset graphs (**~50M+ nodes**).
- Selected and tuned the winning data store for read-heavy query patterns, with targeted index redesign that cut **p95 query latency by ~30%**.
- Added rate limiting and structured error boundaries (circuit breakers) across all APIs, unifying service parity across **4+ independently deployed instances**.

LLM Expert-Routing System (RAG + Vector DB)

- Prototyped an org-wide **agentic AI routing system** using RAG pipelines and vector databases.
- Enabled **semantic expert discovery across 500+ engineers** with automated dual-party notifications.
- Built **continuous self-learning** from resolved queries, eliminating repeat-query overhead.

Cursor & Claude MCP Integration

- Integrated Cursor and Claude MCP into engineering workflows, automating **code review gating** and cursor-rule-driven PR standards.
- Built **semantic Confluence search** to surface relevant documentation during development.
- Reduced team-wide LLM token overhead from ~1.4M to ~910K tokens/week (**~35% reduction**) via prompt engineering, reusable templates, and context pruning.

Dassault Systèmes | Pune, India

May 2021 – Aug 2022

R&D Development Engineer

Stack: ReactJS · Redux · Node.js · MySQL

myMedidata Clinical-Trial Portal

- Built the **ReactJS/Redux frontend** for the global myMedidata patient portal.
- Delivered **multi-locale support across 15+ regional deployments**, serving thousands of clinical-trial participants worldwide.

Node.js REST API Ownership

- Owned end-to-end Node.js REST API integrations on a production-critical healthcare platform, from incident triage and root-cause analysis through hotfix deployment.
- Maintained **SLA-level high availability** across releases.

Test Quality

- Authored unit and integration test suites sustaining >85% code coverage.
- Reduced regression defects by ~20% quarter-over-quarter across release cycles.

Persistent Systems Limited | Nagpur, India

Aug 2018 – May 2021

Software Engineer

Stack: Node.js · Loopback · Java · Redis · Jenkins

Enterprise AppConnect Connectors

- Designed and shipped **7+ production REST connectors** on Node.js/Loopback for Salesforce, AWS DynamoDB, GCS, FileNet, and OracleTaleo.
- Built each with full CRUD, configurable field mapping, **error handling, and retry logic**.

Auth & Security

- Implemented **OAuth2, Cookie-based, and Basic auth flows** across all connector surfaces.
- Handled token refresh, session management, and expiry, meeting enterprise security and compliance standards.

Backend Performance & Reliability

- Profiled connector performance via Grafana/Metis, resolving bottlenecks that **improved throughput by ~25%**.
- Maintained Jenkins CI/CD pipelines with **zero-downtime deployment practices** across connector releases.

Java Integration Layer

- Built **Java-based integration bridges** for FileNet and legacy on-prem systems.
- Enabled bi-directional data sync with the AppConnect middleware platform, **extending connector coverage** to systems without native REST support.

TECHNICAL SKILLS

Languages: JavaScript (Node.js), Python, Java, SQL

Backend & APIs: Node.js, Express, Loopback, Flask, REST APIs, OAuth2, Microservices, API Design, Rate Limiting, Circuit Breakers

AI & LLM: Agentic AI, RAG Pipelines, LangGraph, Vector Databases, Embeddings, Semantic Search, LLM Tool Use, Cursor MCP, Claude MCP, Prompt Engineering, Context Engineering, NumPy, scikit-learn, PySpark, Databricks

Frontend: ReactJS, Redux, HTML5, CSS3

Databases: MongoDB, MySQL, DynamoDB (AWS), Redis, IBM CloudantDB

Cloud & DevOps: AWS (DynamoDB, SNS), IBM Cloud, Docker, OpenShift, Jenkins, CI/CD

Testing: Jest, Cucumber, Mocha/Chai, JUnit, Mockito/Powermockito

Tools: Git, Postman, Jira, VS Code, Grafana, LogDNA, Maven, Dynatrace, Splunk, Catchpoint

System Design: Layered Architecture, Service-Oriented Architecture, Distributed Systems, Graph Data Models, Index Optimization

KEY ACHIEVEMENTS & RECOGNITION

- Site-Level Tech Talk — APS.JSON: Selected to present Autodesk's APS JSON protocol migration to the broader engineering site, **driving org-wide awareness** and accelerating cross-team adoption beyond the immediate product squad.
- First AI-tooling champion on the team — pioneered Cursor & Claude MCP adoption, establishing **reusable prompt templates and cursor rules** that became org-wide engineering assets.
- Connector platform expansion at Persistent Systems — grew AppConnect coverage **from 3 to 10+ enterprise integrations** in under 3 years.

EDUCATION

Lakshmi Narain College of Technology (LNCT) — Bhopal, India

Bachelor of Engineering — Information Technology | CGPA: 8.51 | Graduated 2018